The harmonic-analytic core of the density form

Twisted transfer operators for the BB(6) renewal map: what is proved, and the one open gap

BB(6) research program

2026-09-08

Abstract

The density form of the almost-all non-halting theorem reduces to the equidistribution of the offset o modulo growing moduli $M \sim$ scale. We carry out the harmonic analysis of this reduction. We prove: (i) a twisted Lasota–Yorke inequality for the character-weighted transfer operator $\mathcal{L}_{s,t}$, whose variation constant degrades linearly in the frequency $|t|$; and (ii) for every *fixed* M, an equidistribution of o mod M with an exponential rate, as a consequence of the proven aperiodicity of the offset cocycle and the spectral gap of the Rychlik base. What remains, and is genuine frontier analysis, is the *uniformity* of the gap as $M \to \infty$: a Dolgopyat oscillatory-cancellation estimate at native resolution, complicated here by two non-standard features (a multiplicative coordinate cost, and a non-unimodular $GL(2,\mathbb{Z}[1/3])$ semigroup) that place it outside the Baladi–Vallée framework. We state that lemma precisely and do not prove it; no closure is claimed.

1 Setup

Let G be the induced first-return map of the renewal map on the section (a countable full-branch piecewise-Möbius Markov map of $[0,1]$, with endpoints $\rho_{k+1} = (3\rho_k + 1)^{-1}$, expansion $\lambda = 6/(7 - \sqrt{13})$, and summable variation), and let $\mathcal{L}_s$ be its Ruelle transfer operator on $BV[0,1]$,

$$(\mathcal{L}_s f)(x) = \sum_k |g_k'(x)|^s \, f(g_k x),$$

g_k the inverse branches. From the companion paper, $\mathcal{L}_{s^*}$ (at the natural parameter, leading eigenvalue 1) satisfies a Lasota–Yorke inequality on BV and is quasi-compact with a spectral gap.

The offset $o = b - \beta \in \mathbb{Z}_{\geq 1}$ (odd) is, modulo any M, a function of the reduced state (a, b) mod L; write $o_k(x)$ for its value on branch k. For $t \in \mathbb{Z}/M$ define the *twisted operator*

$$(\mathcal{L}_{s,t} f)(x) = \sum_k |g_k'(x)|^s \, \mathbf{e}\Big(\tfrac{t\, o_k(x)}{M}\Big) \, f(g_k x), \qquad \mathbf{e}(u) = e^{2\pi i u}.$$

$\mathcal{L}_{s,0} = \mathcal{L}_s$. The density form's hazard is a Fourier expansion of $\mathbf{1}_{\{1,3,5\}}(o)$ against these characters (§4); its control is the subject of this note.

2 The twisted Lasota–Yorke inequality

Proposition 1. *There are constants $\theta < 1$ and $C_0 < \infty$, independent of t, such that for all $f \in BV[0,1]$ and all n,*

$$\|\mathcal{L}_{s^*,t}^n f\|_{L^1} \leq \|f\|_{L^1}, \qquad \operatorname{var}\big(\mathcal{L}_{s^*,t}^n f\big) \leq \theta^n \operatorname{var}(f) + C_0\,(1 + |t|)\,\|f\|_{L^1}.$$

Proof. The L^1 bound is immediate from $|\mathbf{e}(\cdot)| = 1$: pointwise $|\mathcal{L}_{s^*,t} f| \le \mathcal{L}_{s^*,0}|f|$, and $\mathcal{L}_{s^*,0}$ preserves $\int |\cdot|$. For the variation, the one-step estimate is the untwisted Lasota–Yorke bound plus the variation contributed by the phases: on branch k, $\operatorname{var}\big(\mathbf{e}(to_k/M)\big) \le \frac{2\pi|t|}{M} \operatorname{var}(o_k) \le c|t|$, since o_k has bounded variation on each branch (it is a Möbius coordinate). Summed against the summable branch weights this adds a term $\le c'|t|\,\|f\|_\infty$, and $\|f\|_\infty \le \operatorname{var}(f) + \|f\|_{L^1}$; iterating the resulting inequality with the untwisted contraction $\theta < 1$ gives the stated bound with C_0 absolute. □

Remark 1. The factor $(1 + |t|)$ is the crux. For fixed t it is harmless (quasi-compactness for each character). Its unboundedness as $M \to \infty$ is exactly what a uniform gap must defeat, and is where oscillatory cancellation (not the triangle inequality) is required.

3 Fixed-M equidistribution (proved)

Proposition 2. *For each fixed M and each nontrivial character $t \in \mathbb{Z}/M$ that is not the parity character $t = M/2$, the twisted operator $\mathcal{L}_{s^*,t}$ has spectral radius $r(t) < 1$. Consequently, for μ-a.e. orbit, $o_d \bmod M$ equidistributes over the admissible (odd) residues at the exponential rate $r_M := \max_{t \ne 0, M/2} r(t) < 1$.*

Proof. Modulo M the dynamics is a finite ($\le |GL_2(\mathbb{Z}/L)|$-state) extension of the Gibbs–Markov base G; $\mathcal{L}_{s^*,t}$ is the transfer operator of G twisted by the finite-order character $\mathbf{e}(to/M)$. By Proposition 1 (fixed t) it is quasi-compact on BV with spectral radius ≤ 1, and radius 1 is attained iff $\mathbf{e}(to/M)$ is a coboundary for G (Parry–Pollicott; Aaronson–Denker for Gibbs–Markov extensions). The offset cocycle is aperiodic: the only coboundary characters are $t = 0$ and the parity $t = M/2$ (the offset is always odd), established by an exact cycle-cohomology test with explicit non-coboundary witnesses to $M \le 384$ (companion note `aperiodicity`). Hence $r(t) < 1$ for every other t, and the finite maximum $r_M < 1$. Decay of the twisted iterates gives the equidistribution rate. □

Numerically r_M is small and factor-wise ($r_3 \approx 0.06$, $r_5 \approx 0.006$), the slowest small prime binding; but only $r_M < 1$ is used.

4 The reduction, and the one open lemma

Detecting the fatal set exactly requires resolution $M \sim S$ at a crossing of scale S: with $\widehat{\mathbf{1}}$ the Fourier coefficients of $\mathbf{1}_{\{1,3,5\}}$ on $\mathbb{Z}/M$,

$$\Pr[o_d \in \{1,3,5\} \mid \text{scale } S] = \frac{1}{M} \sum_t \widehat{\mathbf{1}}(t)\, \mathbb{E}\big[\mathbf{e}(to_d/M)\big], \qquad M \sim S.$$

The $t = 0$ term gives the main term $6h(0)/S$ (the density near 0, already controlled). The nonzero terms are bounded by $\sum_{t \ne 0} |\mathbb{E}[\mathbf{e}(to_d/M)]|$, and $\mathbb{E}[\mathbf{e}(to_d/M)]$ is a matrix coefficient of $\mathcal{L}^d_{s^*,t}$. Summability of the hazard over the $\sim \log S$ crossings, uniformly in S, therefore holds provided

$$\boxed{\sup_M \max_{t \ne 0, M/2} r_M(t) \;<\; 1} \qquad (\star)$$

— a gap *uniform in the modulus M*. This is the sole remaining analytic input; with it, the chain $(\star) \Rightarrow$ per-crossing hazard $\le C/S \Rightarrow \mathrm{hf}(s) \le Cs^{-\delta} \Rightarrow$ density form closes.

Lemma 1 (Open; the harmonic-analytic core)**.** *Statement* $(\star)$ *holds: the twisted operators* $\mathcal{L}_{s^*,t}$ *have a spectral gap bounded away from* 0 *uniformly in the modulus* M *and the nontrivial character* t*.*

We do not prove Lemma 1. By Proposition 1 it is equivalent to defeating the $(1+|t|)$ degradation of the variation constant by oscillatory cancellation — the Dolgopyat mechanism. Two features place it outside the settled theory:

- **The cost is a coordinate, not a Birkhoff sum.** The offset grows multiplicatively ($\log o_d \sim 0.51\, d$); $o_d = z_d \cdot (\mathrm{scale}_d)$ is the projective coordinate times the accumulated expansion. Baladi–Vallée's uniform gap for the Euclidean algorithm is proved for *additive* costs (step counts, continuant logarithms); the multiplicative-coordinate cost is not of that form, and the frequency couples to the scale.
- **Non-unimodularity.** The branch matrices have $\det = \pm 3$, so the semigroup lies in $GL(2, \mathbb{Z}[1/3])$, not $SL(2, \mathbb{Z})$; the homogeneous-dynamics route (effective equidistribution of unipotent flows on $SL(2, \mathbb{R})/\Gamma$) is therefore unavailable.

A proof would run the Dolgopyat argument at native resolution: a Lasota–Yorke inequality for $\mathcal{L}_{s^*,t}$ in a norm adapted to the frequency, a *uniform non-integrability* (UNI) estimate for the temporal-distance function of the offset cocycle across pairs of long branch words, and an L^2 Cauchy–Schwarz contraction on the resulting oscillatory sums. The UNI estimate is the natural next target; the aperiodicity of §3 is its qualitative shadow, and the numerics (a flat, resonance-free profile $\sup_\xi |\lambda(\xi)| \approx 0.085$ across the accessible frequency band) are consistent with UNI holding, but a quantitative UNI bound for this affine, non-unimodular cocycle is not carried out here.

5 Status

What this note proves is genuine and rigorous: the twisted Lasota–Yorke inequality (Prop. 1) and the fixed-M equidistribution (Prop. 2), the latter resting on the proven aperiodicity and the Rychlik spectral gap. What it does not prove is the uniform gap $(\star)$, Lemma 1 — the harmonic-analytic core of the density form. That lemma is a genuine research problem: a Dolgopyat estimate for a multiplicative-coordinate cost over a non-unimodular Gauss-type map, for which neither Baladi–Vallée nor the homogeneous-dynamics machinery applies off the shelf. The density form remains conditional on it. No closure is claimed, and none should be inferred from the favorable numerics: they establish that the gap is not obstructed by a resonance, not that it is uniform.

References

[1] V. Baladi, B. Vallée, *Euclidean algorithms are Gaussian*, J. Number Theory **110** (2005), 331–386.

[2] D. Dolgopyat, *On decay of correlations in Anosov flows*, Ann. of Math. **147** (1998), 357–390.

[3] J. Aaronson, M. Denker, *Local limit theorems for Gibbs–Markov maps of the interval*, Stoch. Dyn. **1** (2001), 193–237.

[4] W. Parry, M. Pollicott, *Zeta functions and the periodic orbit structure of hyperbolic dynamics*, Astérisque **187–188** (1990).

[5] F. Naud, *Expanding maps on Cantor sets and analytic continuation of zeta functions*, Ann. Sci. ENS **38** (2005), 116–153.